

20-52-Q5 -b

Q: (a) CNN

(b) K-means

Solution (b)

① $k=3$

② Sample 1, 2, 3 as centroid C_1, C_2, C_3

③ Iteration 1

Assign all data points to centroid

distance Sample	C_1	C_2	C_3	Assign
1	0	1	1	C_1
2	1	0	1.41	C_2
3	1	1.41	0	C_3
4	6.08	6	5.10	C_3
5	7	7.07	6	C_3
6	6	6.08	5	C_3
7	7.07	6.08	7	C_2
8	6	5	6.08	C_2
9	6.08	5.10	6	C_2

$C_1 : \{1\}$

$C_2 : \{2, 7, 8, 9\}$

$C_3 : \{3, 4, 5, 6\}$

④ New centroid

$$C_1 \rightarrow (0,0)$$

$$C_2 \rightarrow \left(\frac{0+1+0+1}{4}, \frac{1+7+6+6}{4} \right) = (0.5, 5)$$

$$C_3 \rightarrow \left(\frac{1+6+7+6}{4}, \frac{1}{4} \right) = (5, 0.25)$$

⑤ Iteration 2

distance Sample	C1	C2	C3	Assign
1	0	5.02	5.01	C1
2	1	4.03	5.06	C1
3	1	5.02	4.01	C1
4	6.08	6.80	1.25	C3
5	7	8.20	2.02	C3
6	6	7.43	1.03	C3
7	7.07	2.06	7.85	C2
8	6	1.12	7.62	C2
9	6.08	1.12	7.01	C2

$$C_1 = \{1, 2, 3\}$$

$$C_2 = \{7, 8, 9\}$$

$$C_3 = \{4, 5, 6\}$$

$$\textcircled{6} C_1 \rightarrow (0.3333, 0.3333)$$

$$C_2 \rightarrow (0.6667, 6.3333)$$

$$C_3 \rightarrow (6.3333, 0.3333)$$

$\textcircled{7}$ Iteration 3

distance Sample	C1	C2	C3	Assign
1	0.47	6.37	6.34	C1
2	0.75	5.37	6.37	C1
3	0.75	6.34	5.34	C1
4	5.71	7.54	0.75	C3
5	6.67	8.96	0.75	C3
6	5.68	8.28	0.47	C3
7	6.70	0.75	8.54	C2
8	5.68	0.75	8.50	C2
9	5.71	0.47	7.78	C2

$$C_1 = \{1, 2, 3\}$$

$$C_2 = \{7, 8, 9\}$$

$$C_3 = \{4, 5, 6\}$$

⑧ $C_1 \rightarrow (0.3333, 0.3333)$

$C_2 \rightarrow (0.6667, 6.3333)$

$C_3 \rightarrow (6.3333, 0.3333)$

⑨ There is no change in the centroids
and the assignment of samples